Profit Optimization Using Sales Analytics

A Final report for the BDM capstone Project

Submitted by

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Contents

Declaration Statement

I am working on a Project titled "Profit Optimization Using Sales Analytics". I extend my appreciation to **TrendyThreads Pvt. Ltd.** , for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Signature of Candidate:

Amandeep

Name: Amandeep Singh

Date: June 12 2025

1 Executive Summary and Title (285 Words)

TrendyThreads Pvt. Ltd. is a B2C fashion retailer for women that sells through Amazon India. The company has a wide assortment of ethnic and western wear, such as kurtas, tops and dress sets. The company sells to customers in India, using the Fulfilled by Amazon (FBA) model, as well as, the Merchant Fulfilled model.

Even with steady order volume, the company is experiencing a decline in profitability. Certain SKUs report no or low sales, resulting in blocked capital and added carrying costs. The company has high return and cancellation rates (over 11%) which also diminishes revenue and highlights issues with fulfillment or the product itself. Additionally, the company is overly reliant on only two product categories for sales which lowers opportunity for scalability and adds risk.

This project uses exploratory data analysis (EDA), SKU segmentation through clustering, and analysis of performance by fulfillment options to identify inefficiencies. Python, Excel, and Power BI, were utilized to assist in identifying customers buying and returning patterns, as well as delivery performance metrics. The insights developed from the data analysis will inform the management team with key data points that will support decision making processes to optimize inventory management, decrease returns, and offer complementary product offerings that will, hopefully, enable TrendyThreads to become more profitable and allow for long term operational sustainability.

2 Proof of Originality – Data Source

The dataset used in this project, "Unlock Profits with E-commerce Sales Data", was retrieved from Kaggle.

Link:
https://www.kaggle.com/datasets/thedevastator/unlock-profits-with-e-commerce-sales-data

It was originally scraped and published by **Anil Sharma**, with further hosting and access provided via **data.world** at: https://data.world/anilsharma87

The dataset includes overall sales details of an e-commerce fashion retailer selling on Amazon, and has attributes such as Order ID, SKU, Category, Fulfillment Type, Quantity, Order Status, Amount, City/State, and Courier Status that were important for analysis.

3 Meta Data and Descriptive Statistics

3.1 Meta Data

This dataset contains a total of 121000 + records and 24 attributes representing order-level details from Amazon experiences. Each row is a transaction and includes attributes such as Order ID, Date, Status, Fulfilment, Category, SKU, Qty, and Amount.

Qty and Amount are important numerical fields representing product quantity and order sale value, respectively. There are categorical fields for fulfilment methods, product category, and delivery status.

Column Name	Description	Data Type
Order ID	Unique identifier for each order	Object
Date	Order date (standardized to datetime format)	Datetime
Status	Order status: Shipped, Cancelled, Returned, etc.	Object
Fulfilment	Fulfillment channel: Amazon or Merchant	Object
Sales Channel	Marketplace source (Amazon.in)	Object
ship-service-level	Shipping speed (Standard or Expedited)	Object
Style, SKU	Product style and SKU identifier	Object
Category	Product type: Kurta, Set, Top, etc.	Object
Qty	Quantity ordered	Integer
Amount	Revenue from the order	Float
Courier Status	Delivery outcome/status	Object
fulfilled-by	Fulfillment partner (Amazon/Merchant/Estimated)	Object
ship-city/state	Geographic details of delivery	Object

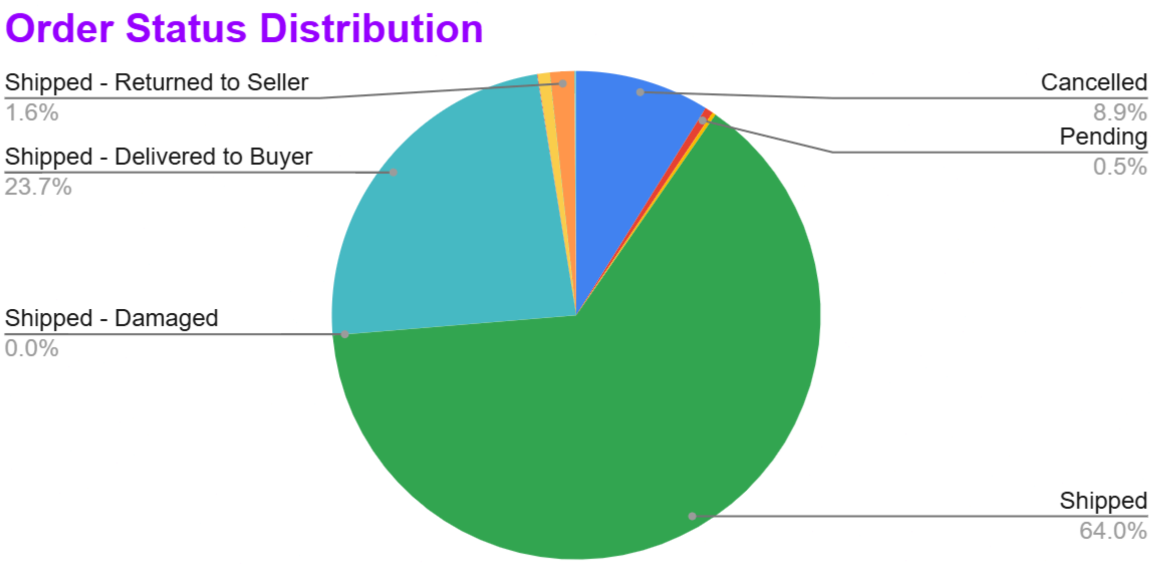
3.2 Descriptive Statistics

- **Average Quantity per Order**: 0.96 units
- **Average Order Value**: ₹648.56
- **Maximum Sale Value**: ₹5,584.00
- **SKUs with Zero Sales**: 19

- **Total Unique SKUs**: 7,157

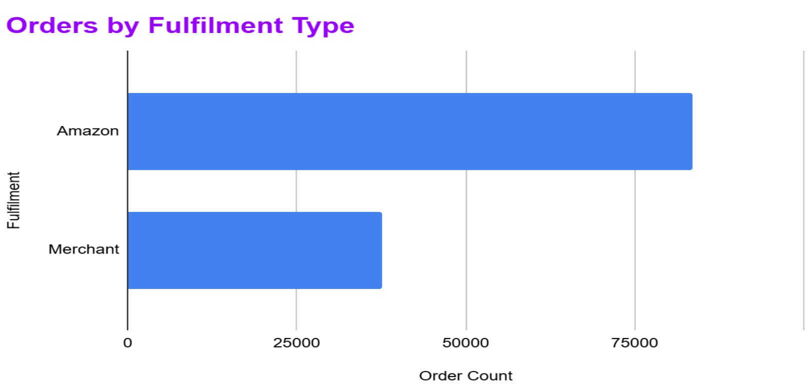
To support further analysis, high-level order distributions are visualized below:

3.2.1 Order Status Distribution



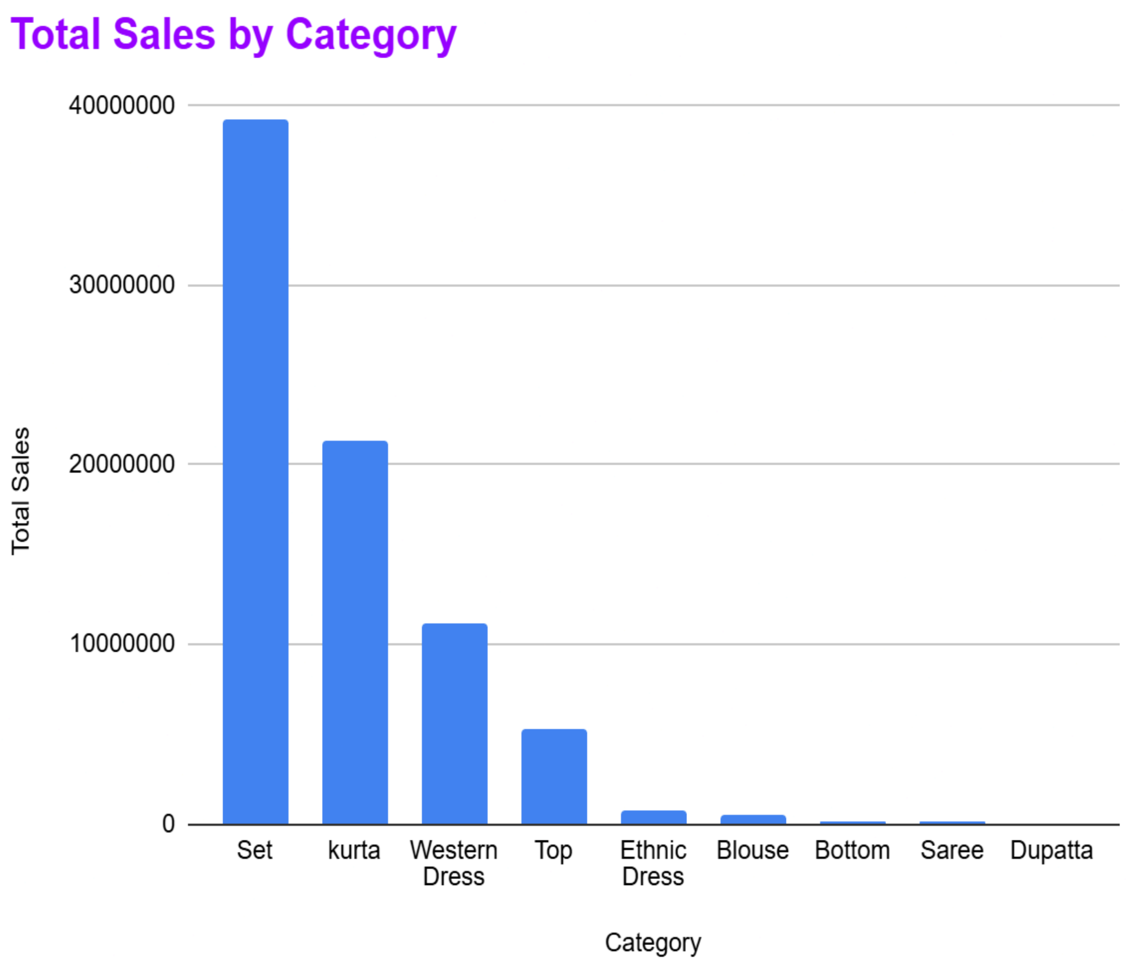
Overall, the dataset has a variety of order outcomes. Most entries have either a status of "Shipped" or "Delivered to Buyer," and many entries have an order status identified as either "Cancelled" or "Returned to Seller," where over 11% of orders are identified with a cancelled or return status. The observation is telling for efficiencies in delivery and customer satisfaction. A pie chart of status frequencies can give a quick visual of the distribution. The fulfillment and returns analysis later in the report further explores these patterns.

3.2.2 Fulfilment Method Split



Orders in the dataset are fulfilled via either Amazon (FBA) or Merchant (promotional (Easy Ship) channels; and in the dataset, 69% of total orders are purportedly Amazon-fulfilled orders and 31% are Merchant orders this is an important metric since fulfillment method can directly impact delivery experience and return rates. The comparative counts of Amazon and Merchant orders are presented in the corresponding bar chart, which will serves as the basis for in-depth group-level comparisons under Section 5.

3.2.3 Sales by Category



Sales are very concentrated in two main product categories: “Set” and “Kurta” together make up greater than 80% of the total order volume and sales and do indicate a very close dependence on a narrow mix of products. A column chart showing total sales by category demonstrates this imbalance. The concentration presents financial risk and restricts customer reach, which buyers will assess in the recommendation section.

3.2.4 Low-Selling SKUs

SKU	Total Sales
BL090-XS	0
BTM003-B-L	0
BTM041-PP-L	0
BTM046-PP-M	0
J0027-SET-XXL	0
J0101-DR-A-XL	0
J0324-CD-L	0
JNE3486-KR-L	0
JNE3486-KR-XXL	0
JNE3748-KR-S	0
JNE3783-KR-XS	0
JNE3788-KR-L	0
JNE3910-KR-M	0
MEN5015-KR-S	0
PJNE3439-KR-N-6XL	0
SET301-KR-PP-XL	0
SET306-KR-PP-S	0
SET361-KR-NP-L	0

19 SKUs showed no sales during the observation period. These SKUs could be dead stock, new products/unlisted products, or problems with inventory. Before considering restocking, promoting or removing, a decision has to be made on these SKUs

4 Detailed Explanation of Analysis Process / Method

A structured and data-driven process was used to solve the main business problems for TrendyThreads Pvt. Ltd. and include stagnant inventory, high return rates, poor fulfillment efficiencies, and category overdependency. This process blends quantitative and qualitative methods to produce actionable insights.

4.1 Data Cleaning and Preparation

Original dataset underwent thorough data cleaning in order to give accurate and consistent analysis. Important fields with missing values such as Amount, Fulfilment and Courier Status were accounted through either imputation or deletion. Dates were converted to datetime format and unnecessary fields were removed. This resulted in a cleaned dataset of around 121,000 rows and 20+ fields.

```python
import pandas as pd

# Load the uploaded file
df = pd.read_csv("Amazon Sale Report.csv")

# Drop unnecessary columns
df = df.drop(columns=["Unnamed: 22", "promotion-ids"], errors='ignore')

# Drop rows where Amount or currency is missing
df = df.dropna(subset=['Amount', 'currency'])

# Convert Date to datetime format
df['Date'] = pd.to_datetime(df['Date'], format='%m-%d-%y', errors='coerce')

# Fill missing 'fulfilled-by' with 'Fulfilment'
df['fulfilled-by'] = df['fulfilled-by'].fillna(df['Fulfilment'])

# Fill location-related missing values with 'Unknown'
for col in ['ship-city', 'ship-state', 'ship-country', 'ship-postal-code']:
    df[col] = df[col].fillna('Unknown')

# Fill missing 'Courier Status'
df['Courier Status'] = df['Courier Status'].fillna('Unknown')

# Save cleaned file
df.to_csv("Amazon_Sale_Report_Cleaned.csv", index=False)
print("Cleaned CSV file saved!")
```

4.2 Exploratory Data Analysis (EDA)

Objective: The objective was to get a holistic picture of the dataset with respect to trends and distributions in order amounts, item counts, fulfillment status, and order outcomes (shipped, cancelled, returned) to help set the boundaries of inefficiencies and potential areas of opportunity.

Method: Exploratory Data Analysis was done using Google sheets, Python (Pandas, Matplotlib) and Excel, where summary statistics were calculated for sales amount, item quantities, status, etc. I used pivot tables and charts to visualize key metrics such as return rate, cancellation rate, and fulfillment performance.

Why I chose it: Exploratory Data Analysis is a core process in which to understand any dataset. Using EDA provided me the ability to quickly identify trends hidden in the data, identify key outliers, and illuminate patterns, especially in a transactional dataset that is rich in information. I wanted to makeUTE of as many tools as I could, while still providing myself a means of flexibility in a number of ways.

4.3 SKU Performance and Segmentation

Objective: The goal was to evaluate how each SKU performs on the various dimensions of revenue, returns, and fulfillment to determine which SKUs are high-performing, and which are problematic (i.e. dead-inventory).

Method: The SKUs were grouped using clustering logic (K-Means in Python) and prioritized by total sales, return rate, and fulfillment method, marking SKUs with no or low sales as dead inventory.

Why I chose it: Segmentation gives actionable insights at the product level. This enables the organization to manage product and SKU level objectives of product sales and to assign SKU level decisions like whether to clarify the inventory, restock, or promote. This would also help category planning and margin optimization.

4.4 Fulfillment Comparison Analysis

Objective: Assess if fulfillment mode - Amazon Fulfilled, or Merchant Fulfilled - influenced outcomes such as return, cancellation, or successful delivery rates.

Method: Orders were analyzed on the basis of fulfillment type, and grouped statistical comparisons were developed. The two fulfillment paths were visually compared using stacked column graphs and summary tables in Google Sheets.

Why I chose it: Fulfillment experience affects customer satisfaction and operational cost - what works better is the question. Comparing the two models also tells us if we need to spend more time optimizing logistics, or if one is more efficient for high performing SKUs.

4.5 Category-Level Revenue & Risk Exposure

Objective: To evaluate which product categories drive revenue and whether the organization's sales are overly dependent on a small number of categories and to assess business risk and potential opportunities for diversification.

Method: Sales data was organized by product category. Then for each category total revenue, total number of orders and average sale price were calculated. A Pareto chart and a series of column graphs were created to illustrate category sales.

Why I picked it: Understandably, concentration analysis is very important in long-term strategic planning. Having identified that two categories (Kurta and Set) generate greater than 80% of revenue raises risk awareness to reassess diversification or hedging strategies in the other categories.

4.6 Tools Used

Google Sheets & Excel: Pivot tables, bar and pie charts, basic filtering and visualization.

Python: For data cleaning, grouping, clustering, and visualization.

4.7 Link to Supporting Analysis Sheet

To ensure transparency of the analytical process, full operational analysis including pivot tables, calculated fields, and visualization can be accessed through the following link.

Amazon_Sale_Report_Cleaned

5 Results and Findings

This section gives the main findings of the data analysis performed on TrendyThreads Pvt. Ltd. Amazon sales data. Each finding relates to a specific business challenge previously identified in the proposal, including poor inventory efficiency, returns, category performance, and fulfillment inefficiencies.

5.1 SKU Performance and Dead Inventory

Objective: To identify SKUs that are contributing minimally or nothing to overall sales, and to what extent dead inventory is present.

Methodology: Sales data was aggregated by SKU using Google Sheets (and Excel Pivot tables). Total revenue per SKU was calculated, and reported in a Treemap chart.

Key Findings: 19 SKUs had zero sales, and several others contributed under ₹100. These are wasted/terrible inventory, and take up space and capital.

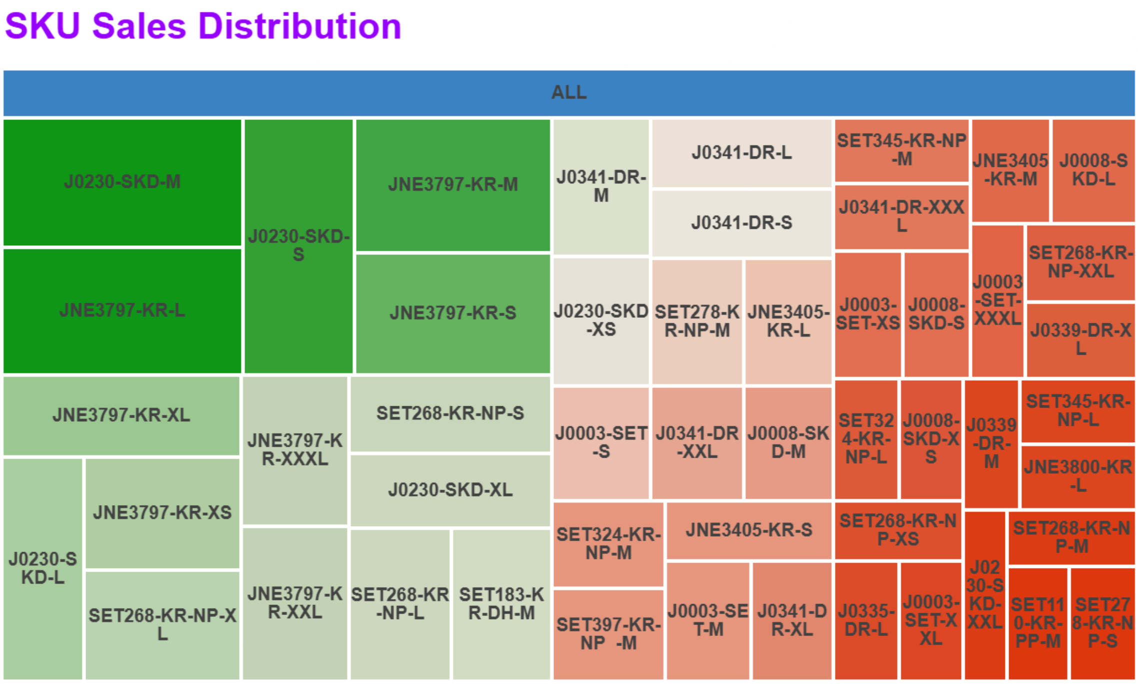


Figure 5.1: Contribution of SKUs to Total Revenue (many show zero or minimal sales)

5.2 Order Status Breakdown and Customer Delivery Issues

Objective: To measure the extent of cancelled and returned orders, while identifying trends or categories with issues.

Methodology: Order status was analyzed and accounted for using a pivot table technique - counting each outcome (Delivered, Cancelled, Returned). The order status was shown and visualized in a pie chart.

Key Findings: Overall approximately 10% or more of orders were cancelled or returned. Merchant-fulfilled orders and segments such as Kurta and Set have the higher return ratio as notable conclusions.

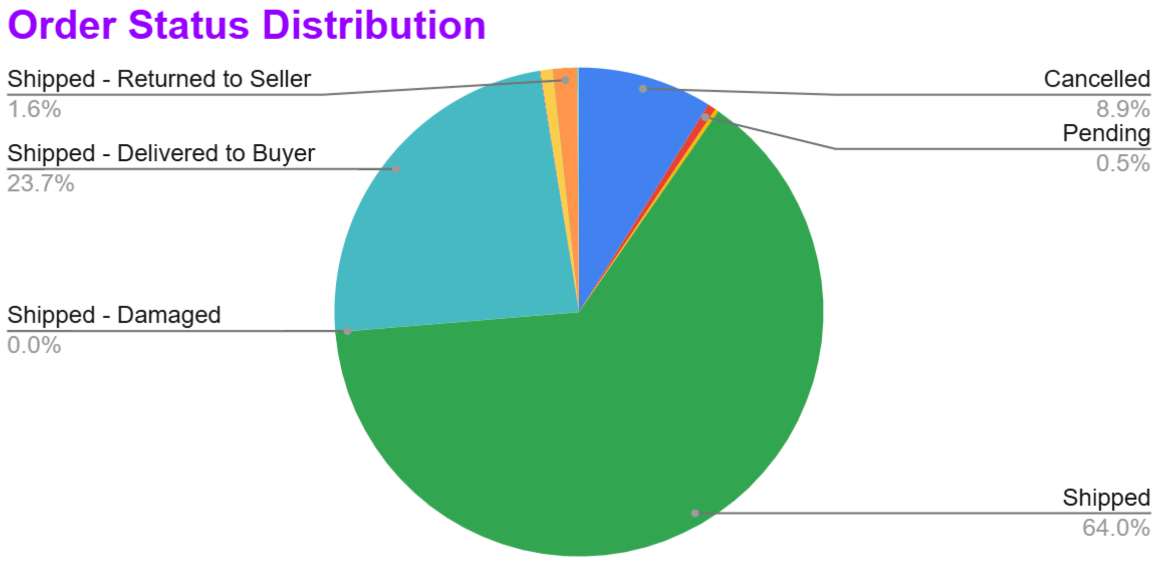


Figure 5.2: Proportion of Delivered vs Cancelled and Returned Orders

5.3 Fulfilment Channel Comparison

Objective: Compare the performance of Amazon Fulfilment vs Merchant Fulfilment in terms of delivery success and order returns.

Methodology: The orders were segregated by fulfilment type and then the statuses for the different fulfilments were grouped and plotted. The distribution was also plotted in 100% stacked column and bar charts.

Key Findings: Amazon-fulfilled orders have a statistically significantly better delivery success and lower cancellations than Merchant. The Merchant-fulfilled orders actually comprised a small portion of the revenue compared to returns/cancellations.

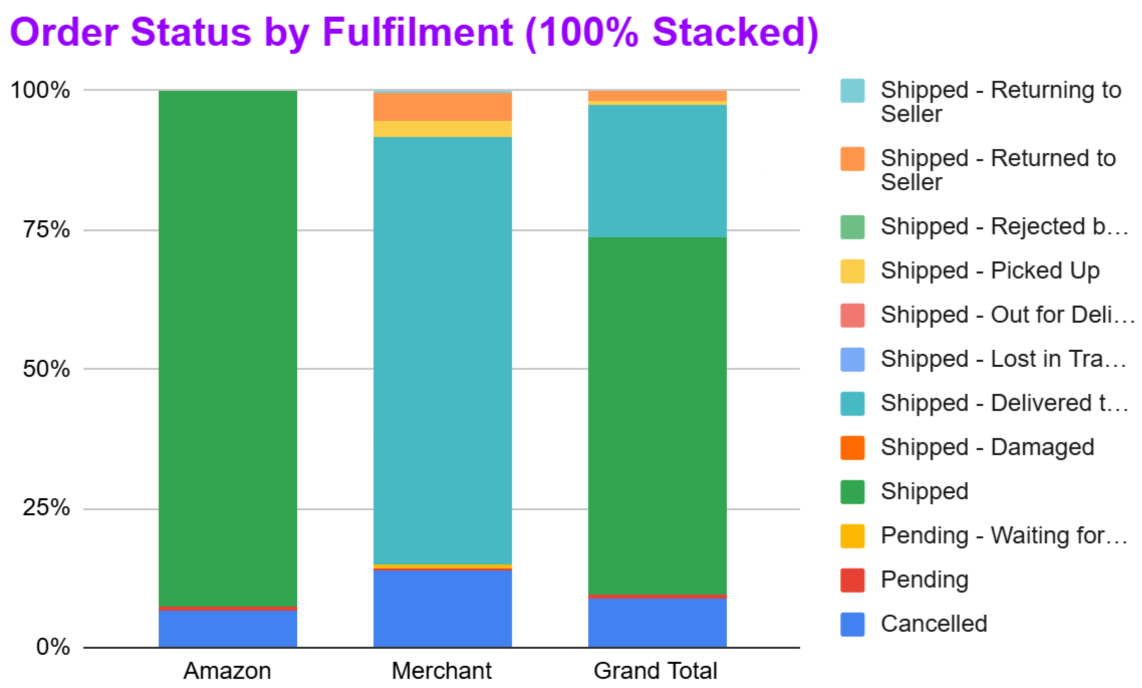


Figure 5.3: 100% Stacked Column Chart showing proportion of order statuses for Amazon vs Merchant fulfilment.

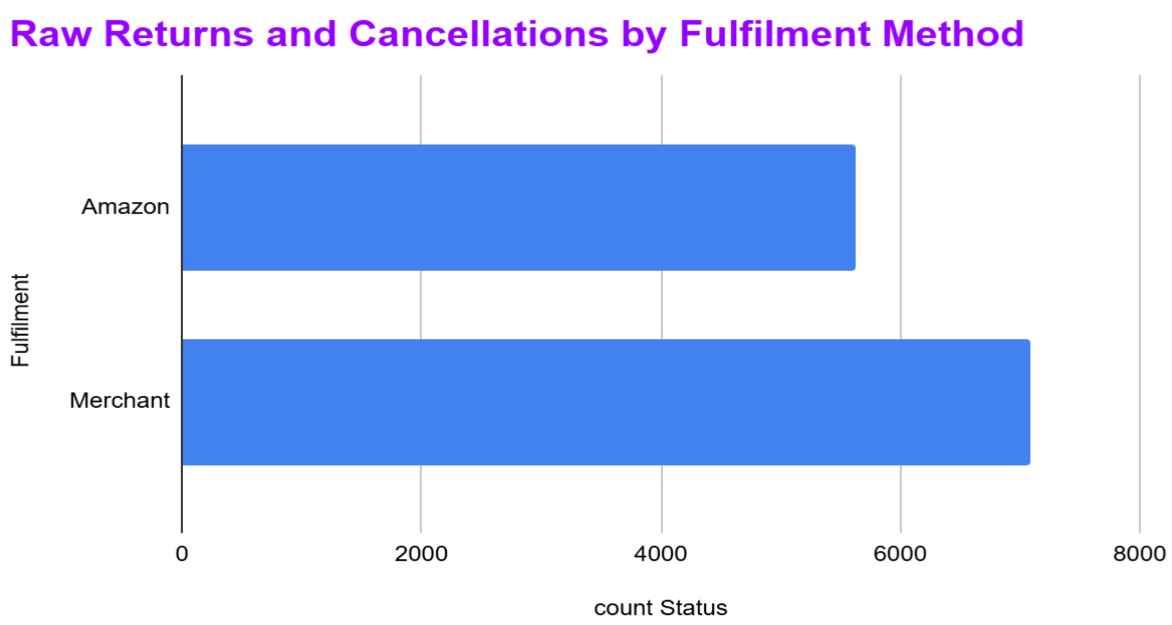


Figure 5.3b:Bar chart comparing the raw count of returned and cancelled orders by fulfilment type.

5.4 Category-Level Concentration Risk

Objective: To assess how dependent the business is on certain product categories and the business risks of that dependence.

Methodology: Total revenue aggregated by category. A Pareto chart and column graph demonstrated distribution and indicated dominance trends.

Key Findings: Kurta and Set categories combined contribute to over 80% of total revenue. There is business risk if demand in those categories fades.

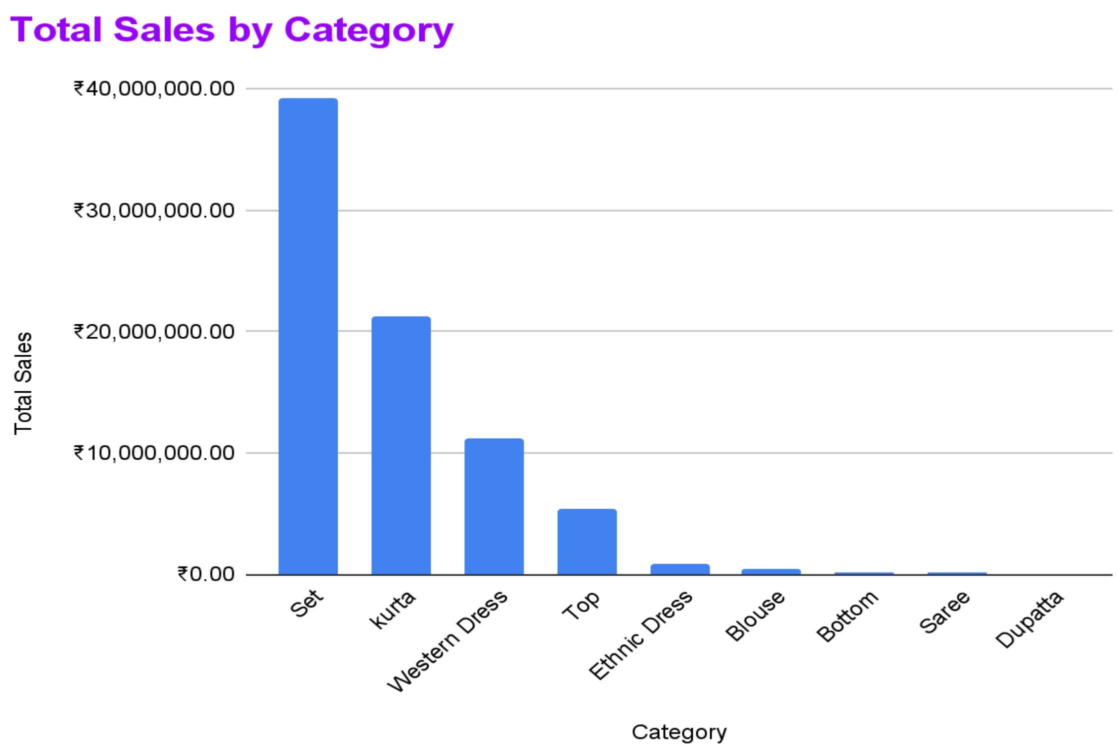


Figure 5.4: Column chart showing sales by product category. Kurta and Set dominate the revenue share.

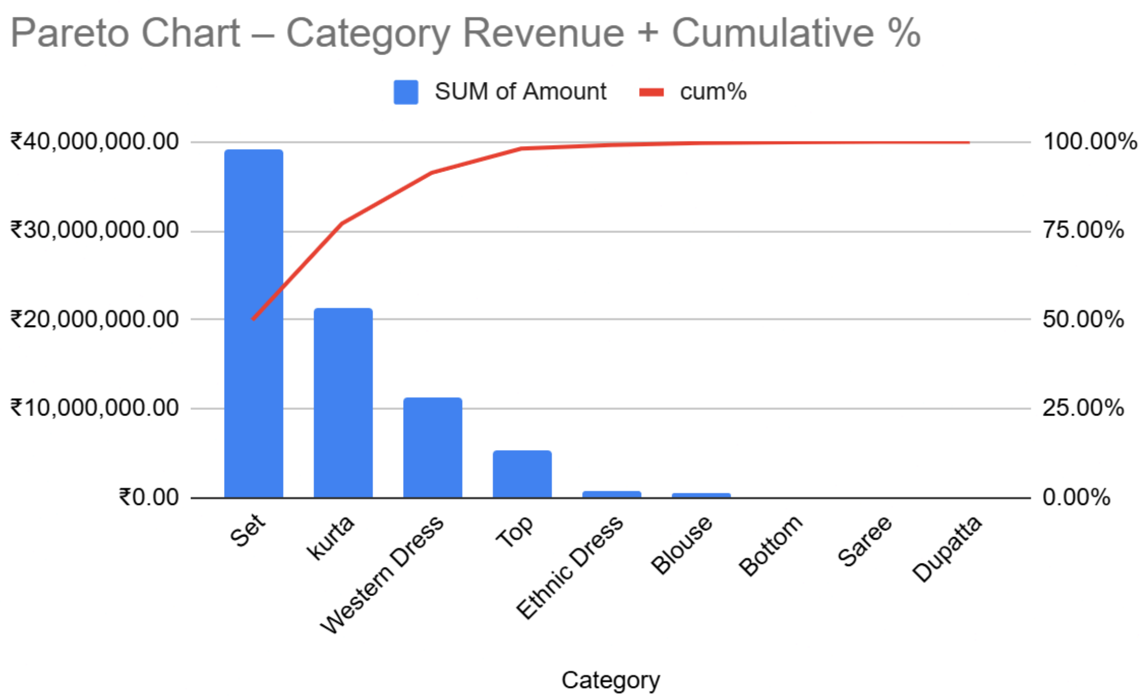


Figure 5.4b: Pareto chart visualizing the 80/20 rule. A few categories generate the majority of revenue.

5.5 Daily Revenue and Sales Pattern

Objective: To examine the daily variation in revenue and determine patterns involved with demand and marketing strategy.

Methodology: Date-based sales were grouped into a line chart in Google Sheets to show revenue patterns over each time a sales trend was analyzed.

Key Findings: Sales fluctuated significantly over different dates suggesting promotional spikes and or unstable demand. A more consistent approach might smooth out these variations in revenue.

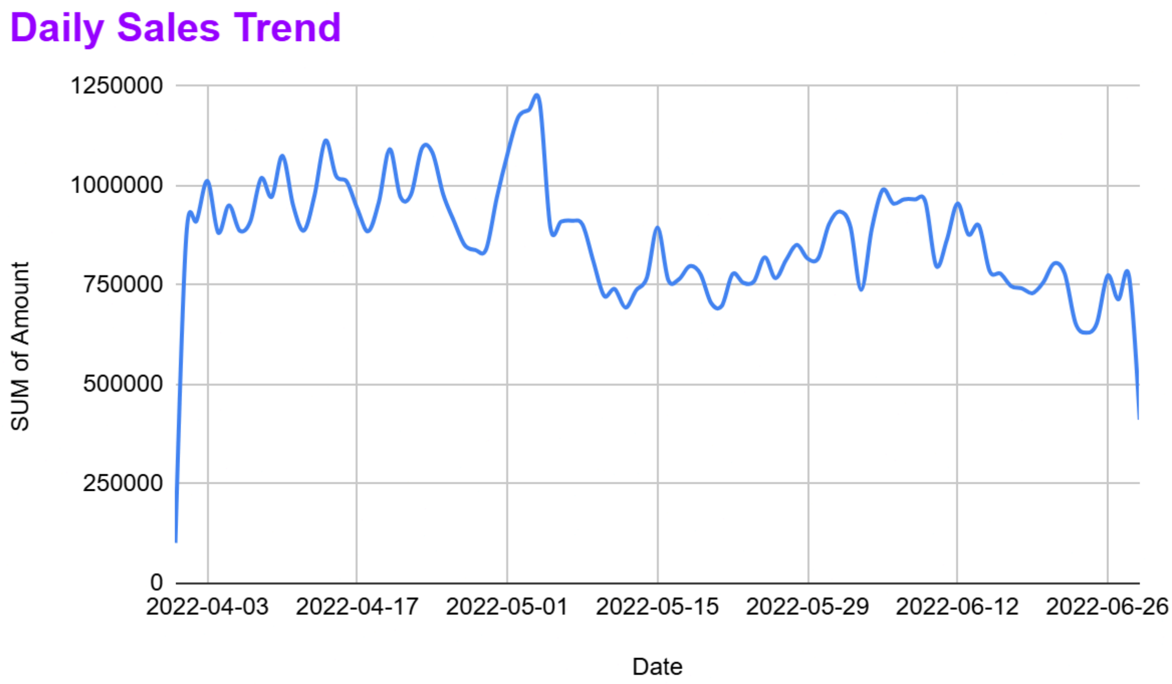


Figure 5.5: Line chart representing daily sales trend over time, highlighting peaks and troughs in revenue.

5.6 Geographic Distribution

Objective: To find the top 10 regional states with effective sales leverage through revenue generation and sales volume through amounts sold.

Methodology: Sales amounts were categorized by states and visualization was performed by way of a horizontal bar chart centered on the ten best states.

Results: Urbanized states like Maharashtra, Karnataka, and Delhi are strong sales states-rightfully so, indicating focuses for marketing and regional inventory allocation.

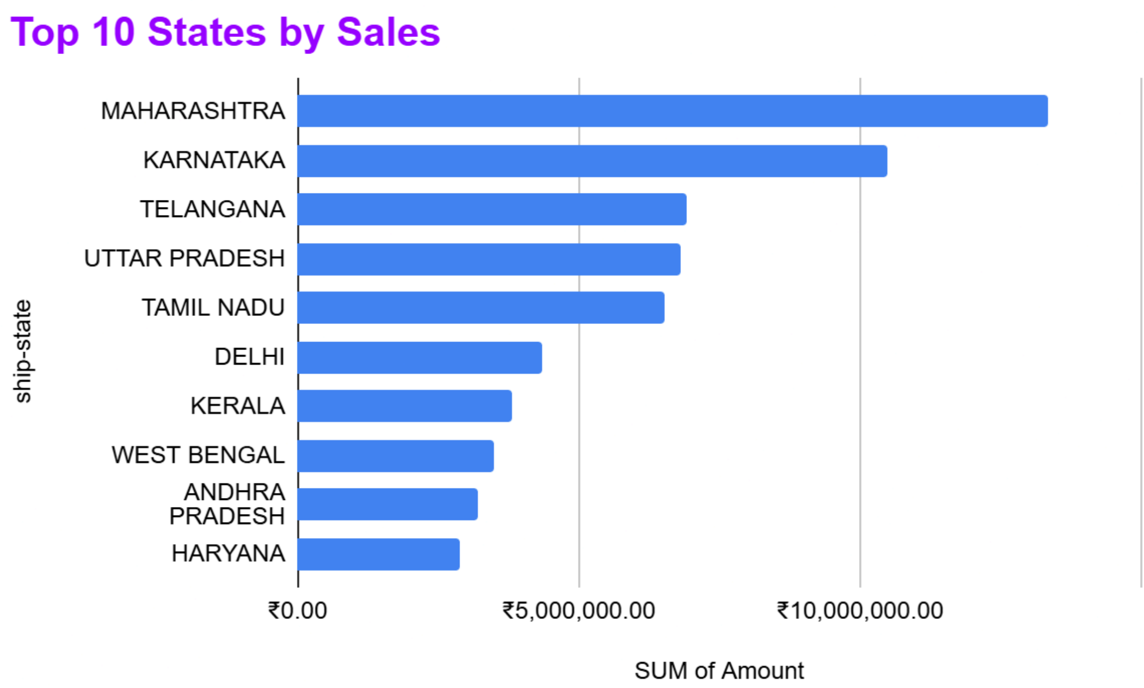


Figure 5.6: Revenue Contribution by State – Top 10 Regions

5.7 SKU Bubble Analysis

Objective: To view SKU-level results across dimensions- quantity sold, gross, and category.

Methodology: We created a bubble chart which used total sales, quantity and category metrics. This showed SKUs plotted to show their overall contribution.

Key Findings: The position of bubbles helped ascertain SKUs which sold in volume, but were not profitable, and those which excelled across measures.

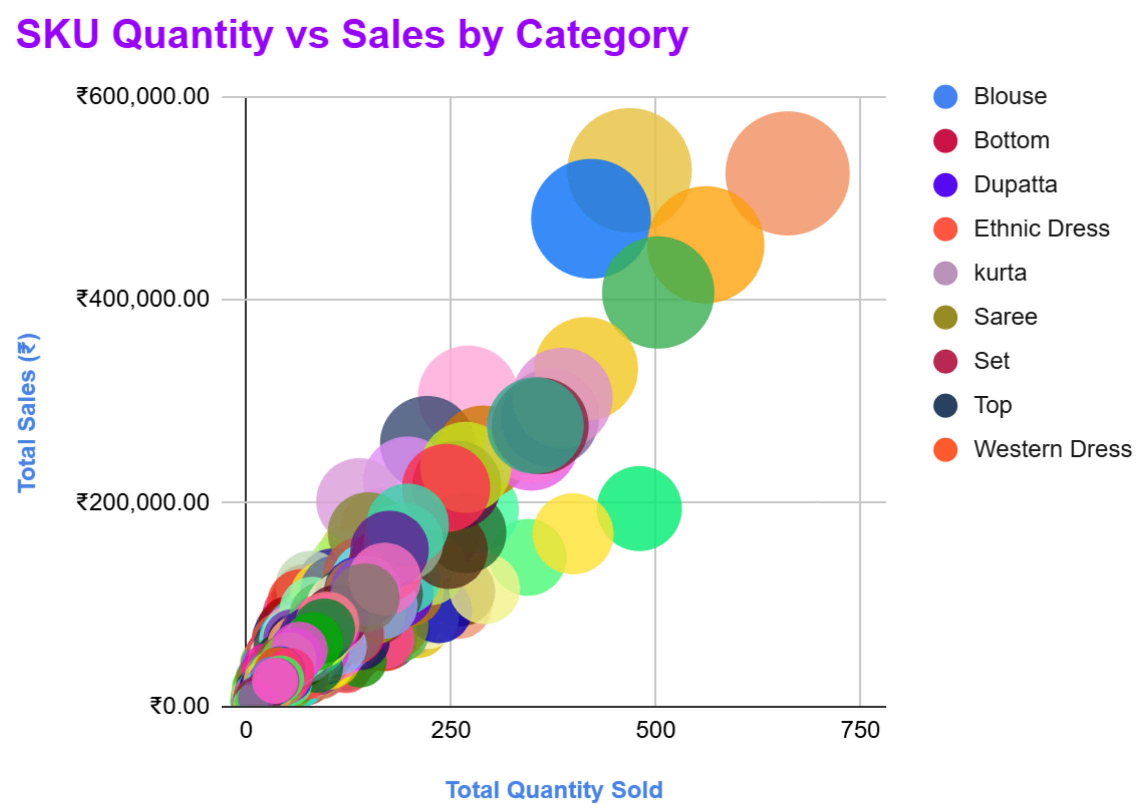


Figure 5.7: Bubble chart showing multi-metric SKU analysis based on sales, quantity, and product category.

6 Interpretation of Results and Recommendations

6.1 Interpretation of Results

The analysis revealed significant performance gaps aligned with the business issues stated above:

6.1.1 Problem 1 (Unsold/Low-Selling SKUs):

The SKU-level analysis suggested several SKUs have zero or negligible sales. This stagnant inventory represents a blockage of capital and an unneeded storage cost. The treemap visual (Figure 5.1) demonstrates the inconsequential revenue generation relatively speaking of so many SKUs as well.

6.1.2 Problem 2 (High Order Cancellations and Returns):

The order status analysis (Figure 5.2) identified that more than 14% of orders were either cancelled or returned. A subsequent comparison of fulfilment type (Figure 5.3) indicated that Merchant-fulfilled orders were having a disproportionate effect on this issue. This suggests potential delays in logistics, or expectations between normative and actual delivery were not aligned.

6.1.3 Problem 3 (Sales Dependence on Few Categories):

The category-level revenue analysis (Figures 5.4 and 5.4b) indicate that just two categories of product, Kurta and Set, are generating the vast majority of revenue. Reduced diversification of product categories places the business at risk for loss in revenue, should demand become seasonal or concentrated in trend shifts.

6.1.4 Other Insights:

The analysis of temporal and geographic sales trends (Figures 5.5 and 5.6) provide clarity on the within day demand density on each day, in addition to identifying a concentration of sales in three regions. This provides intelligence on key assumptions, and provides guidance on important marketing, stocking and fulfilment strategies.

6.2 Recommendations

To address the above challenges, we propose the following data-driven recommendations:-

6.2.1. Inventory Rationalization:

- Liquidate SKUs with no sales, or very low sales, to allow for the allocation of renew capital and shelf space to larger ticket items.

6.2.2. Enhancing Customer Experience:

- Improve our product images, sizing guides and product descriptions to improve matched expectations and ultimately achieve lower levels of returns. Use Fulfilled-by-Amazon (FBA) for faster, reliable delivery.

6.2.3. Diversification Process:

- Launch new fashion categories or slow moving products bundled with our top sellers to test cross selling/multiple selling.

6.2.4. Data-Driven Regional Approach:

- Utilize our data to evaluate our states' top performers with local offers (state and possibly local) and strategically account for placing regional inventory hubs to lower delivery time and reduce delivery costs.

6.2.5. Daily Monitoring of Sales:

- Set up dashboards to monitor daily sales which will help identify a rapid drop in sales, or a spike in sales from when running demand-optimized campaigns.

7 Presentation and Legibility of the Report

The report has been prepared with clarity, formatting, and ease of understanding in mind. Each major section has a format and structure, including being numbered as follows: introduction, problem statements, data exploration, findings, and recommendations. There is limited technical language in the report; where technical language is used, it is explained so as to make sense to non-technical readers.

The charts and graphs provided in Section 5 have been placed in a different colour block colour scheme; and further, each figure is labelled with a descriptive caption (e.g., Figure 5.2: Order Status per Business Sector). This makes the report more visually manageable and navigation-friendly. The type of charts was decided upon by the type of data, including treemaps, pie charts, line charts, and stacked bar charts. These charts aided in the description of the data and did not detract from the data or contents of the report.

In Google Sheets, data was cleaned, transformed, and charts were built; ultimately, Google Sheets allows for a more transparent and reproducible review of the steps taken. The report is now grammar checked, formatted consistently, and designed according to academic and professional presentation standards, so that it is appropriate for both assessment and presentation purposes with stakeholders.